

COURSE TITLE : POLYMER ADHESIVES
COURSE CODE : 6095
COURSE CATEGORY : E
PERIODS/WEEK : 4
PERIODS/SEMESTER: 60/6
CREDITS : 4

TIME SCHEDULE

Module	Topics	Period
1	Introduction to adhesives Surface preparation for adhesive bonding	15
2	Adhesive materials such as . Animal glue, casein and mixed protein adhesives, starch based adhesives, natural rubber adhesives.	15
3	Butyl rubber and poly iso butylenes adhesive. Poly vinyl alcohol for adhesives. Nitrile rubber adhesives. polyurethane-isocyanate based adhesives-development-reasons for the effectiveness types and uses-(A-D) Test methods	15
4	Styrene butadiene adhesives, Neoprene based solvent and latex adhesives, Phenolic resin adhesives ,Epoxy resin adhesives	15
TOTAL		60

COURSE OUTCOME

Sl.	GO	Student will be able to
1	1	To understand the general characteristics of adhesives
2	2	To know the structure, properties and application of Natural adhesives
3	3	To know the preparation ,properties and application of synthetic adhesives such as Butyl rubber, Polyvinyl alcohol, Nitrile rubber adhesives
4	4	To know the other adhesives such as SBR, Neoprene, Phenolic and Epoxy resins.
4	4	To understand typical compounding materials, Formulation and Testing of Adhesives.

SPECIFIC OUTCOME

MODULE I INTRODUCTION TO ADHESIVES

1.1.0 To understand the general characteristics of and classification of adhesives.

- 1.1.1. State the advantages of adhesive bonding.
- 1.1.2. Give the history of adhesives. .
- 1.1.3. Explain the types of adhesives

- 1.2.4. State the origin of adhesives.
- 1.2.5. Explain Hybrids and Coupling agents. .
- 1.2.6. Explain Matching adhesive to Adherend.
- 1.2.7. Explain the factors for designing polymer adhesives.
- 1.2.8. Explain the factors considering surface preparation for adhesive bonding.

MODULE II ADHESIVE MATERIALS

2.2.0 To know the preparation ,properties and application of animal glue, casein and mixed protein adhesive, starch based adhesives, natural rubber adhesives.

- 2.1.1. Introduction of animal glue.
- 2.1.2. Explain the chemistry of animal glue.
- 2.1.3. Brief study of amino acids in animal glue table.
- 2.1.4. Explain the properties of animal glue and its application.
- 2.1.5. Describe the history of casein.
- 2.1.6. Describe the chemical and physical properties of casein.
- 2.1.7. Illustrate the applications of casein solution and adhesives.
- 2.1.8. Describe the additives for casein and protein blend glues.
- 2.1.9. Uses of casein glues.
- 2.1.10. Introduction of starch glue.
- 2.1.11. Describe the modifications of starches.
- 2.1.12. Describe the effect of additives in starch.
- 2.1.13. Describe the application areas of starch glues.
- 2.1.14. Introduction of natural rubber adhesives.
- 2.1.15. Specification of SMR graded natural rubber.
- 2.1.16. Preservatives for natural rubber.
- 2.1.17. Describe the manufacturing of natural rubber grafted with methyl methacrylate.
- 2.1.18. Describe the formulation of latex adhesives.
- 2.1.19. Describe the types of NR based adhesives.

MODULE III ADHESIVE MATERIALS

3.1.0 Introduction of butyl rubber and its basic properties.

- 3.1.1 Describe the properties of polyisobutylene and halogenated butyl rubber.
- 3.1.2 Describe the formulating and processing of butyl rubber lattices.
- 3.1.3 Applications of butyl rubber latices.
- 3.1.4 Introduction of polyvinyl alcohol for adhesives.
- 3.1.5 Describe the manufacture of polyvinyl alcohol adhesive..
- 3.1.6 Introduction to nitrile rubber adhesives.
- 3.1.7 Preparation of nitrile rubber adhesive.
- 3.1.8 Types of compounding ingredients for nitrile rubber adhesives.
- 3.1.9 Applications of nitrile rubber adhesives
- 3.1.10 Polyurethane and isocyanate based adhesives
- 3.1.11 Describe the development of PU adhesives
- 3.1. 12 Reasons for the effectiveness of PU adhesive
- 3.1.13 Types and use of polyurethane adhesive system(method-A-D)
- 3.1.14 Describe the test methods of adhesives

MODULE IV ADHESIVE MATERIALS

4.1.0 Introduction to styrene butadiene adhesive.

- 4.1.1 Describe the SBR latices.
- 4.2.2 Describe the compounding ingredients in SBR latices.
- 4.1.3 List the different compounding ingredients with Examples .
- 4.1.4 Introduction of neoprene based adhesives.
- 4.1.5 Describe the effect of polymer structure
- 4.1.6 Describe the additives in neoprene adhesives.
- 4.1.7 List out the applications of neoprene adhesives .
- 4.1.8 Introduction of Phenolic resin adhesives..
- 4.1.9 Describe the manufacture of phenolic resin adhesives.
- 4.1.10 Explain the use of phenolic resins in abrasives ,coatings, and laminating
- 4.1.11 Describe the epoxy resin using adhesives.
- 4.1.12 Describe the curing agents used in epoxy adhesives.

CONTENT DETAILS

MODULE I

Adhesives-characteristics—advantage-history-types-origin.Hybrids-coupling agents. Adherends-matching adhesives to adherend. Factors for design adhesives-surface preparation.

MODULE-II

Adhesive materials-preparation-properties-application of Animal glue Casein-mixed protein adhesive, Starch adhesive, Natural rubber adhesive. Animal glue-introduction-chemistry-study of amino acids-animal glue table-properties-applications. Casein-history-chemical and physical properties-applications. Natural rubber-SMR grade-preservatives-manufacturing-grafting with Methacrylate formulation-types

MODULE III

Properties of Butyl-isobutyl-halogenated Butyl- applications .Poly vinyl alcohol –introduction-manufacture .Nitrilerubber-preparation-introduction-compounding ingredients-applications. polyurethane and isocyanate based adhesives-development-reasons for effectiveness-types –use-A-D test methods-peeling, lap shear, tensile detachment-torsional , fracture, cleavage, pull off, blister tests

MODULE IV

SBR-introduction-compounding ingredients-examples.Neoprene –effect of polymer structure-additives – applications-.Phenol formaldehyde-introduction-manufacture uses in coatings-abrasives laminating. Epoxy resin- resin used adhesives, curing agents

REFERENCE

1. Handbook of adhesives Third edition Edited by IRVING SKEIST
2. PLASTIC MATERIALS EDITED BY AJ BRYDSON
3. WIKIPEDIA